Linear growth factor of density perturbations in FRW cosmologies.

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Abstract

In this notes we show how to derive and solve numerically the second order differential equation for the linear growth factor of density perturbations, $D(a)$, with an arbitrary dark energy equation of state $w(a)$.

1 The equations

According to Peebles (1980), the time evolution for matter perturbations is ruled by the following equation

$$\frac{\partial^2\delta}{\partial t^2} + 2H\frac{\partial\delta}{\partial t} - \frac{3}{2}H\Omega_m\delta = 0, \tag{1}$$

where $\delta(\vec{x},t)+1 = \rho(\vec{x},t)/\bar{\rho}(t)$ is the density contrast, $H(t)$ is the Hubble parameter and $\Omega_m(t)$ is the matter density parameter.

The linear growth factor $D(t)$ is defined in terms of the evolution of the density perturbations as $\delta(\vec{x},t) = D(t)\delta(\vec{x},t_0)$, being t_0 a particular time which defines the normalization, i.e. $D(t_0) = 1$. Using this, the differential equation for $D(t)$ is

$$\frac{d^2D}{dt^2} + 2H\frac{dD}{dt} - \frac{3}{2}H\Omega_m D = 0 \tag{2}$$

At this point it is convenient to change the time variable to the scale factor. If we do the change $t \to a$, the derivate operators change as

$$\begin{aligned}\frac{d}{dt} &= Ha\frac{d}{da}\\ \frac{d^2}{dt^2} &= (Ha)^2\frac{d^2}{da^2} + Ha\left(a\frac{dH}{da} + H\right)\frac{d}{da}\end{aligned}$$

Using this in Eq. (2) we obtain

$$\frac{d^2D}{da^2} + \left(\frac{dH}{da}\frac{1}{H} + \frac{3}{a}\right)\frac{dD}{da} - \frac{3}{2}\frac{\Omega_m}{a^2}D = 0 \tag{3}$$

In a matter-dominated Universe $D = a$, so we introduce the variable $G(a) = D(a)/a$ in order to remove that dependency at early times. With this new variable, Eq. (3) takes the form

$$\frac{d^2G}{da^2} + \frac{1}{a}\left(5 + \frac{d\ln H}{d\ln a}\right)\frac{dG}{da} + \frac{1}{a^2}\left(3 - \frac{3}{2}\Omega_m + \frac{d\ln H}{d\ln a}\right)G = 0 \tag{4}$$

In order to avoid the a^{-1} and a^{-2} factors in the second and third terms, respectively, it is convenient to use $\ln a$ as time variable, so

$$\begin{aligned}\frac{d}{da} &= \frac{1}{a}\frac{d}{d\ln a}\\ \frac{d^2}{da^2} &= \frac{1}{a^2}\frac{d^2}{d\ln a^2} - \frac{1}{a^2}\frac{d}{d\ln a}\end{aligned}$$

With this last change, the equation takes the final form

$$\frac{d^2G}{d\ln a^2} + \left(4 + \frac{d\ln H}{d\ln a}\right)\frac{dG}{d\ln a} + \left(3 - \frac{3}{2}\Omega_m + \frac{d\ln H}{d\ln a}\right)G = 0. \tag{5}$$

The Hubble parameter can be expressed in terms of the density parameters of the different components of the Universe. If we consider radiation (γ), matter ($m = dm + b$), curvature (K) and dark energy (DE), we have

$$H = H_0\sqrt{\Omega_\gamma^0 a^{-4} + \Omega_m^0 a^{-3} + \Omega_K^0 a^{-2} + \Omega_{DE}^0 e^{-3\int_0^{\ln a}[1+w(a')]d\ln a'}} \tag{6}$$

where the superscript 0 indicates values today. With this, the logarithmic derivate of the logarithm of $H(a)$ can be expressed as

$$\frac{d\ln H}{d\ln a} = -\frac{1}{2}\left[4\Omega_\gamma + 3\Omega_m + 2\Omega_K + 3(1+w(a))\Omega_{DE}\right] \tag{7}$$

where the density parameters as a function of the scale factor are given by

$$\begin{aligned}
\Omega_\gamma &= \frac{\Omega_\gamma^0}{a^4}\frac{H_0^2}{H^2} \\
\Omega_m &= \frac{\Omega_m^0}{a^3}\frac{H_0^2}{H^2} \\
\Omega_K &= \frac{\Omega_K^0}{a^2}\frac{H_0^2}{H^2} \\
\Omega_{DE} &= \frac{\Omega_{DE}^0}{e^{3\int_0^{\ln a}[1+w(a')]d\ln a'}}\frac{H_0^2}{H^2}
\end{aligned}$$

Using Eq. (7) in Eq. (5), and considering that $\sum_{i=\gamma,m,K,DE}\Omega_i = 1$, we obtain the final expression

$$\frac{d^2G}{d\ln a^2} + \left[\frac{5}{2} + \frac{\Omega_K}{2} - \frac{\Omega_\gamma}{2} - \frac{3}{2}w(a)\Omega_{DE})\right]\frac{dG}{d\ln a} + \left[2\Omega_K + \Omega_\gamma + \frac{3}{2}(1-w(a))\Omega_{DE}\right]G = 0 \tag{8}$$

By solving Eq. (8), solutions for $G(a)$ and its derivate can be found. These two quantities are easily related with the linear growth factor and its derivate:

$$D(a) = \frac{G(a)}{a} \tag{9}$$

$$\frac{d\ln D}{d\ln a} = \frac{1}{G}\frac{dG}{d\ln a} + 1 = f(\Omega). \tag{10}$$

2 The code

The code is written in C and uses some rutines from Numerical Recipies (Press et al., 2002). The package contains the following files:

- `allvars.c`: Provide instances for all global variables.
- `allvars.h`: Declares all global variables.
- `cosmology.c`: Cosmological functions and subroutines.
- `io.c`: Read input file and set up global variables.
- `main.c`: Main program.
- `numrec.c`: Numerical Recipies routines to solve ODE and to perform integrations.
- `proto.h`: Functions and subroutines prototypes.

2.1 Input file

The user must define the following variables/options in the input file:

- `UseTab_wEOS`: Set to 1 for a tabulated time dependent dark energy equation of state $w(a)$ or set to 0 for a linear behaviour.
- `wEOSFile`: File with the tabulated $w(a)$. Ignored if `UseTab_wEOS` $= 0$.
- `OutputFile`: Name of the file for the output.
- `OmegaRadiation0`: Radiation density at $z = 0$.
- `OmegaMatter0`: Matter density at $z = 0$.
- `OmegaDarkEnergy0`: Dark energy density at $z = 0$.
- `OmegaCurvature0`: Curvature density at $z = 0$.
- `Hubble0`: Hubble parameter at $z = 0$, in [Mpc km^{-1} s^{-1}].
- `w0EOS` and `waEOS`: Dark energy equation of state of the form $w(a) = w_0 + w_a(1 - a)$. Ignored if `UseTab_wEOS` $= 1$.
- `InitialRedshift`: Initial redshift for the tabulated output.
- `FinalRedshift`: Final redshift for the tabulated output.
- `Nbins`: Number of tabulated values in the output.

If `UseTab_wEOS` $= 1$, the user must provide an `ASCII` table with the values for the scale factor, a, and the dark matter equation of state, $w(a)$, preferentially equally spaced in $\log(a)$.

2.2 Output file

The output is an `ASCII` table with the following tabulated quantities, equally spaced in $\log(a)$.

- `Column` #1: Cosmological scale factor, a.
- `Column` #2: Redshift, z.
- `Column` #3: Linear growth factor, $D(a)$, normalized so that $D(a = 1) = 1$.
- `Column` #4: Cosmological growth rate, $f(\Omega) = d \ln D / d \ln a$.
- `Column` #5: Radiation density parameter, $\Omega_\gamma(a)$.
- `Column` #6: Matter density parameter, $\Omega_m(a)$.
- `Column` #7: Curvature density parameter, $\Omega_K(a)$.
- `Column` #8: Dark energy density parameter, $\Omega_{DE}(a)$.
- `Column` #9: Hubble parameter, $H(a)$.

References

Peebles P. J. E., 1980, The large-scale structure of the universe

Press W. H., Teukolsky S. A., Vetterling W. T., Flannery B. P., 2002, Numerical recipes in C++ : the art of scientific computing